

Saral Banker

Full Stack Engineer · AI Application Developer · Founding Engineer Candidate

Ahmedabad, Gujarat, India · saralbanker1@gmail.com · +91 9016990136 github.com/saralbanker https://orivon-re.vercel.app/

PROFESSIONAL SUMMARY

Full Stack and AI Application Engineer who architects and ships complete, production-grade systems — from PostgreSQL schema and REST API design through to React frontend and containerised deployment. Independently designed and built Neuro-Zenith (NeuroDashboard): a 70k+ LOC modular AI productivity platform spanning a full RAG pipeline, Socket.IO real-time collaboration, async job processing (BullMQ/Redis), multi-provider LLM routing (OpenRouter, Gemini, local Qwen 2.5 Instruct), RBAC auth, Sentry error tracking, Prometheus metrics, and a 4-pipeline GitHub Actions CI/CD system. Delivered a second production system as a paid contract — a financial operations platform that replaced manual Excel billing and physical collection calls for a 220-unit rental business, enforcing automated late-fee penalties and saving an estimated 40+ hours per monthly billing cycle. Experienced in AI model evaluation across GPT-4, Claude, Gemini, DeepSeek, Qwen, Kimi, Gemma, and MiniMax. At my best with hard problems, full ownership, and the autonomy to ship.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python, SQL, Java, Kotlin, HTML, CSS
Frontend	React 18, Next.js, Vite, Tailwind CSS, shadcn/ui, Radix UI, Framer Motion, TanStack React Query, Zustand, Tiptap, Recharts, FullCalendar
Backend	Node.js, Express 4, FastAPI, REST API Design, Socket.IO 4, SSE
AI & LLMs	RAG Pipelines · pgvector (HNSW) · Semantic Search · BGE-base-576M Embeddings (Sentence-Transformers) · OpenRouter API · Gemini API · Local LLM Inference (Qwen 2.5 Instruct) · Prompt Engineering · Structured Output Parsing (Zod) · AI Workflow Automation · Multi-Provider LLM Integration · AI Model Evaluation (GPT-4, Claude, Gemini, DeepSeek, Qwen, Kimi, Gemma, MiniMax)
Databases	PostgreSQL (raw pg driver, 44 migrations), Supabase, Redis 7, MySQL, SQLite, pgvector
Infrastructure	Docker, Docker Compose, GitHub Actions (CI + Staging CD + Prod CD + Terraform IaC), GHCR, Terraform, Supabase CLI
Architecture	Modular Monolith + Intelligence Microservice · BullMQ Async Job Processing · DB Polling Queue (SKIP LOCKED) · Event-Driven (Redis Pub/Sub) · Multi-Tenant RBAC · JWT Auth · Rate Limiting · Audit Logging · Software Requirements Engineering · PRD Writing · System Architecture Design
Observability	Sentry (@sentry/node), Prometheus (prom-client), Pino structured logging
Mobile	Kotlin, XML, SQLite (Android)
Tools	Zod, Vitest, Jest 29, Supertest, Postman, Git, Cisco Packet Tracer

EXPERIENCE

Contract Full Stack Engineer

Jan 2025 – Mar 2026

Financial Operations Platform (Shade Ledger) · Confidential Client · Ahmedabad, Gujarat

- Owned the complete project lifecycle for a rental property financial operations platform serving 220 units — replaced a workflow where the client manually entered data in Excel and physically called each of 220 vendors every month, with no enforceable penalty for late fees due to face-to-face relationship dependency.
- Built a recurring billing engine with automated PDF invoice generation (react-pdf), WhatsApp Business API direct delivery, and structured late-fee penalty logic — eliminating ad-hoc fee waivers and recovering previously lost penalty revenue on every billing cycle.
- Engineered a bulk Excel import pipeline with row-level schema validation and error reporting, reducing monthly tenant data entry from a multi-hour manual process to a single structured file upload.
- Built a maintenance management module for tracking repair requests, vendor assignments, and resolution status across all 220 units.
- Designed an immutable financial audit log capturing all mutations (payments, voids, penalty charges, adjustments) with actor identity, timestamp, and before/after state — replacing informal verbal agreements with a traceable, queryable system of record.

- Delivered a complete, documented, production-ready PostgreSQL-based financial system as a Rs.60,000 paid contract engagement, independently managing requirements through to client handoff; eliminated an estimated 40+ hours of manual billing and collection work per monthly cycle.

Stack: React, Next.js, TypeScript, Node.js, PostgreSQL, Supabase, react-pdf, WhatsApp Business API

PROJECTS

Neuro-Zenith (NeuroDashboard) [70k+ LOC]

2025 – 2026

github.com/saralbanker/neuro-zenith · Role: Founder, Lead Architect, Lead Developer

Independently designed and shipped a production-grade modular AI platform across a full-stack TypeScript monolith with an attached Python intelligence microservice — architected from first principles across 400+ source files, 115 directories, and 44 database migrations.

- **Core Modules:** Knowledge Hub (AI knowledge management, semantic document retrieval, embeddings, vector search, AI summaries, action-item extraction, note-to-task conversion) · Notes System with real-time collaboration · Tasks & Action Items · AI Insights Engine · Workspace Management · Analytics Dashboard · Activity Tracking · Document Processing Pipeline · Calendar (FullCalendar) · Admin & Billing (Stripe).
- **Architecture:** Designed a Modular Monolith + Intelligence Microservice architecture: Node.js/Express core handling auth, REST routes, WebSocket, and job orchestration, with a FastAPI Python sidecar (uvicorn, port 8000) owning all CPU-bound embedding and semantic inference — communicating via internal HTTP, sharing a single PostgreSQL/pgvector source of truth.
- **RAG Pipeline:** Built a full RAG pipeline: document ingestion → text extraction → chunking → BGE-base-576M embedding generation (Sentence-Transformers, Python) → pgvector HNSW index storage → hybrid semantic + keyword retrieval — serving the `/api/search/semantic` endpoint.
- **Async Processing:** Designed a dual async processing system: BullMQ/Redis event-driven worker (`aiWorker.ts`) handling AI jobs (generate-embedding, note-summary, note-action-items, note-to-tasks, file-process, email dispatch) and a DB polling queue (`SELECT FOR UPDATE SKIP LOCKED`, 2s interval) for transactional jobs — decoupling all heavy processing from API response latency.
- **LLM Routing:** Integrated multi-provider LLM routing via OpenRouter API (cloud — GPT-4, Gemini, and others), Gemini API, and local Qwen 2.5 Instruct (offline mode) behind a unified `AIProvider` abstraction layer — enabling model switching, cost-aware routing, and fallback logic without changes to business logic.
- **Auth & RBAC:** Implemented JWT authentication with a 3-layer RBAC system: `authMiddleware.ts` (GoTrue token validation), `workspaceMiddleware.ts` (membership scoping), and `projectMiddleware.ts` (project-level isolation) — enforced via `roleMiddleware.ts` and `permissionMiddleware.ts` on all protected API surfaces.
- **Real-Time:** Built real-time collaboration using Socket.IO 4 across two namespaces (`/collab`, `/notes`) backed by Redis pub/sub for multi-process event broadcasting — supporting concurrent document editing and live notification dispatch.
- **Observability:** Configured full production observability: Sentry (`@sentry/node`) for error telemetry, Prometheus (`prom-client`, `/metrics` endpoint) for runtime metrics, and Pino structured HTTP logging — giving the system complete crash and performance visibility.
- **CI/CD & Deployment:** Deployed via a 4-pipeline GitHub Actions system: PR CI (Node 18/20 matrix, unit + integration tests), Staging CD (`pg_dump` DB backup + migrations + smoke tests + auto-rollback), Prod CD (Docker build + push to GHCR), and Terraform IaC pipeline — running against a Docker Compose stack (Node server, Redis 7, Python service).
- **Database & Testing:** Designed a 44-migration PostgreSQL schema (raw pg driver, no ORM) with multi-tenant workspace isolation, Supabase RLS policies, soft-delete, and full audit logging — supported by 30 backend test files (Jest 29 + Supertest) covering auth, queue, semantic search, WebSocket, and AI service layers; 6 frontend test files (Vitest + Testing Library).

Stack: React 18, Vite 7, TypeScript, Node.js 20, Express 4, FastAPI, Python 3.14, PostgreSQL, Supabase, Redis 7, BullMQ 5, pgvector, Socket.IO 4, OpenRouter, Gemini API, Qwen 2.5 Instruct, BGE-base-576M, Sentence-Transformers, Docker, GitHub Actions, Terraform, Sentry, Prometheus, Pino, Zod, TanStack Query, shadcn/ui, Framer Motion, Tiptap, FullCalendar, Stripe

Smart Parking Management System

2025

github.com/saralbanker/smart-parking

- Built a real-time parking occupancy system with live slot status broadcasting, online slot booking, reservation management, and an admin capacity dashboard.
- Designed optimistic concurrency control for simultaneous check-in/check-out requests, preventing double-booking under concurrent load.

Stack: React, Node.js, Express, Supabase, Tailwind CSS, TypeScript, WebSocket

Carbon Compass

2025

github.com/saralbanker/carbon-compass

- Designed a personal emissions tracking platform with activity logging across transport, energy, and food categories, an emissions modelling engine, and a time-series sustainability reporting dashboard.

Stack: React, TypeScript, Vite, Tailwind CSS, shadcn/ui, Chart.js

Employee Management System

2024

Built a full HR management system for employee tracking, workforce administration, department management, and attendance records.

Stack: HTML, CSS, JavaScript, MySQL

Home Service Platform

2025

Developed an Android application for home service booking, customer management, and appointment scheduling with a service provider matching flow.

Stack: Kotlin, XML, SQLite (Android)

Tax Calculator System

2024

Built a GST and income tax computation tool with multi-slab logic and financial summary output.

Stack: Java, React

Steam Boiler Company Website

2024

Redesigned and modernised a steam boiler company's corporate web presence, delivering a clean responsive site to replace an outdated static HTML page.

Stack: HTML, CSS, JavaScript

ADDITIONAL EXPERIENCE & CAPABILITIES

- AI Model Evaluation — hands-on comparative evaluation of GPT-4, Claude (Anthropic), Gemini, DeepSeek, Qwen, Kimi, Gemma, and MiniMax across reasoning, code generation, and instruction-following tasks.
- Prompt Engineering Systems — design and iteration of system prompts, few-shot examples, chain-of-thought frameworks, and custom prompt generation pipelines for production AI features.
- Software Requirements Engineering — authored Product Requirement Documents (PRDs), system design specifications, and technical documentation across multiple projects.
- Knowledge System Design — designed domain ontologies, chunking strategies, retrieval evaluation frameworks, and document workflow pipelines for AI-powered knowledge platforms.
- Workflow Automation Design — designed semi-automated billing, notification, and document processing workflows replacing fully manual operational processes.
- Cisco Packet Tracer — designed and simulated WLAN network topologies for academic infrastructure planning project.
- AI-Assisted Development Workflows — integrated AI coding assistants into development workflow for code review, architecture validation, and documentation generation.

EDUCATION

Diploma in Computer Engineering
LJ Polytechnic, Ahmadabad

May 2026

CGPA 8.36 / 10 · Top 10% of class

CERTIFICATIONS

IBM	Python for Data Science, AI & Development · Generative AI: Prompt Engineering Basics · Getting Started with Git and GitHub
Google	Foundations of Cybersecurity · Agile Project Management

Jatin Basantani

AI-Integrated Full-Stack Developer · Software Engineer

Ahmedabad, Gujarat, India • +91 70160 41857 • jatinbasantani861@gmail.com • github.com/jatin-861

PROFESSIONAL SUMMARY

Diploma in Computer Engineering graduate (CGPA 8.61/10, LJ Polytechnic · LJ University) with hands-on experience building AI-integrated full-stack web applications using React, Next.js, Tailwind CSS, and Supabase. Proficient in integrating frontier AI models (Gemini, GPT-4, Claude) through custom prompt engineering and agent-style workflows. Passionate about AI systems, automation, and research tooling. Google- and Kaggle-certified. Targeting Software Engineering, AI/ML, and Product Engineering roles at Big Tech and AI-first organisations.

EDUCATION

Diploma in Computer Engineering

2022 – 2025

LJ Polytechnic · LJ University, Ahmedabad

CGPA: 8.61 / 10

- Awarded Best Mentor / Moderator — recognised for mentoring 5+ peers on full-stack dev and AI integration over 18 months
- Core coursework: Data Structures, DBMS, Computer Networks, OS, OOP
- Designed and simulated LAN/WAN topologies using Cisco Packet Tracer for networking coursework

PROJECTS

Vaida.ai — AI-Powered Vedic Medicine Advisor

Next.js · Tailwind CSS · Gemini API · Custom Prompt Engineering

- Built a natural-language health advisor mapping user symptoms to Ayurvedic remedies via Gemini API with a custom system prompt enforcing safe, evidence-aligned responses
- Crafted structured system prompt instructing the model to act as a certified Ayurvedic practitioner — limiting scope to natural remedies and flagging serious conditions for professional care
- Developed responsive Next.js / React frontend with real-time AI-streamed responses and clean UX for non-technical users

MediSphere — Hospital Management System

React · Node.js · Express · SQL · Role-Based Auth

- Architected full-stack hospital platform with 10+ modules: patient registration, appointment scheduling, billing, and doctor dashboards
- Implemented RBAC with three distinct roles (Admin, Doctor, Patient), each with scoped permissions and dedicated UI flows
- Built RESTful API layer in Node.js/Express with normalised SQL schema for patients, appointments, and billing records

Knowledge Hub — AI Research & Note-Taking Platform

Next.js · React · Supabase · shadcn/ui · Tailwind · AI Integration

- Building AI-augmented knowledge management tool enabling users to capture, organise, and query research notes via natural-language AI interactions
- Integrated frontier AI models for contextual summarisation, auto-tagging, and Q&A; over user content using custom prompt pipelines
- Leveraged Supabase (Postgres) for real-time sync, auth, and structured content storage; designed data model around notes, tags, and AI metadata

Smart Study Lab — AI Study Assistant

React · Next.js · AI API Integration · Tailwind CSS

- Developed intelligent study platform with AI-driven flashcard generation, quiz creation, and adaptive tutoring via frontier LLM APIs
- Designed prompt engineering pipeline to auto-generate MCQs and concept summaries from user-pasted material

Smart Money Hub — Personal Finance Tracker

React · Supabase · Tailwind CSS · Data Visualisation

- Built full-stack finance dashboard for tracking income, expenses, and budgets with real-time Supabase backend and interactive chart visualisations
- Integrated AI-assisted insights to surface spending patterns and budget recommendations from transaction history

Bulk Object Remover — AI Image Processing Tool

React · AI Vision API · REST Integration

- Built browser-based tool for batch object removal across multiple images simultaneously using AI vision API
- Designed drag-and-drop UI with async processing pipeline handling concurrent image uploads and API calls

Additional Projects — Salon Management · Invoice Generator · Food Manufacturing Site

React · Node.js · Tailwind CSS · Client Deliverables

- Salon Management System: full booking, staff, and inventory management web app presented to a real client stakeholder as a functional prototype
- Firm Invoice Generator: automated invoice creation with PDF export built for a business use case
- B2B/B2C Food Manufacturing Site: UI/UX enhancement of a commercial food-sector website with fully responsive design

TECHNICAL SKILLS

Languages	JavaScript / TypeScript, Python, Java, C/C++, SQL
Frontend	React, Next.js, Tailwind CSS, shadcn/ui, HTML5/CSS3, Flutter/React Native
Backend & DB	Node.js, Express.js, Supabase (Postgres), REST API design & consumption
AI / ML	Gemini API, OpenAI GPT-4, Claude API, Prompt Engineering, Agent Architecture, AI Evaluation
Tools	Git, GitHub, VS Code, Cisco Packet Tracer, Agile / Scrum

CERTIFICATIONS & RECOGNITION

- Google Career Certificates — Foundations of Cybersecurity, Digital Marketing, Data Analytics
- Kaggle Learn — Intro to Machine Learning, Pandas, Python
- Agile & Scrum Fundamentals — course completion certificate
- Best Mentor / Moderator Award — LJ Polytechnic (sustained peer mentorship in web dev & AI, 18+ months)
- Letter of Appreciation — issued by institution for academic contributions and leadership
- Multiple course completion certificates across AI, web development, and computer networking